Akhil Velluva

Bioinformatician / Postdoctoral Researcher

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PROFESSIONAL SUMMARY

Bioinformatics scientist and data engineer with 12+ years of experience developing scalable, reproducible computational workflows for multi-omics and translational research. Experienced in designing HPC/GPU-based pipelines using Nextflow, Snakemake, Python, R and containerized environments to process and integrate genomic, bulk and single-cell RNA-seq, spatial transcriptomic and clinical datasets. Develop automated analytical workflows, patient-level molecular reports and reusable data resources supporting biomarker discovery, patient stratification and therapeutic research. Experienced in multi-omics data integration, quality-controlled data processing, AI-agent-assisted biomedical data interpretation and mentoring researchers within interdisciplinary scientific and clinical teams.

CORE SKILLS

Multi-Omics Data Engineering	Genomics, genome-wide association studies (GWAS), statistical genetics, bulk RNA-seq, single-cell RNA-seq, spatial transcriptomics (10x Visium/Xenium), multi-omics integration, data QC, harmonization and large-scale biological data processing.
Scientific Workflow Engineering	Nextflow, Snakemake, Linux/Bash, HPC/LSF, GPU computing, Docker, Singularity, reproducible pipeline development, workflow automation
Programming & Data Analysis	Python, R/Bioconductor, SQL/MySQL, Git/GitHub
AI & Machine Learning	scikit-learn, PyTorch, MOFA/MUVI, clinical outcome prediction, biomarker discovery, patient stratification, AI-agent-assisted data interpretation and workflow automation
Translational & Computational Biology	Cancer genomics, immunotherapy, tumor microenvironment, biomarker discovery, clinical cohorts, target discovery

PROFESSIONAL EXPERIENCE

Nov 2023 - Present

Bioinformatician / Postdoctoral Researcher | *HI-TRON*

Helmholtz Institute for Translational Oncology, Mainz, Germany

- Design and develop scalable HPC/GPU-based computational pipelines for processing and analysis of large-scale genomic, transcriptomic, spatial and clinical datasets.
- Integrate multi-omics datasets, including genomic, transcriptomic, spatial and clinical information, to support biomarker discovery, therapeutic target identification and patient stratification.
- Develop reusable and automated R/Python-based workflows for data processing, QC, harmonization, analysis and downstream reporting.
- Develop automated reporting workflows that translate somatic variant findings into patient-level molecular oncology reports aligned with AMP/ASCO/CAP guidelines for clinical interpretation and reporting.
- Apply machine-learning and patient-projection approaches to large cancer cohorts for outcome prediction, biomarker discovery and identification of molecularly similar patient populations.
- Develop computational approaches for tumor microenvironment and immune-cell profiling in cancer immunotherapy research.
- Collaborate closely with experimental scientists, clinicians and computational researchers to define analytical requirements and translate complex biological datasets into reproducible data products.

- Evaluate and apply AI-agent-assisted approaches for scientific data interpretation and workflow automation.

Aug 2019 - Aug 2023

Scientific Assistant / PhD Candidate | *University of Leipzig, Germany*

- Analyzed the *Steller sea cow* ancient genome to profile mechanisms of Arctic adaptation.
- Developed online resources to help clinical genetics teams select suitable tissues for transcriptome analysis.
- Analyzed RNA-seq and exome data to investigate obesity mechanisms.
- Developed and implemented computational approaches to understand mechanisms underlying genetic diseases.

Jun 2017 - Jul 2019

Senior Scientific Officer | *SRM-DBT Platform for Advanced Life Science Technologies, Chennai, India*

- Analyzed NGS data generated by the core facility and delivered customized analysis solutions to customers.
- Trained junior scientific officers in NGS data analysis and supported facility-level analytical workflows.

Aug 2015 - Mar 2017

Research Associate | *ICAR - Central Institute of Brackishwater Aquaculture, Chennai, India*

- Metagenomic and whole-genome analysis of *Vibrio* and *Pseudomonas* species; developed a variant database/tool for *V. parahaemolyticus*.

Mar 2014 - Jul 2015

Research Fellow | *ICAR - Central Plantation Crops Research Institute, Kasaragod, India*

- Applied machine learning to protein and functional genomics problems in coconut (DELLA, translation initiation factors, MAP kinase domains); developed Smrna, a standalone pipeline for plant miRNA mining and annotation.

Jan 2014 - Mar 2014

Research Project Trainee | *ICAR - Central Plantation Crops Research Institute, Kasaragod, India*

- Performed transcriptome analysis and annotation of red palm weevil (NGS) and homology modelling of coconut ent-copalyl diphosphate synthase.

ACADEMIC CREDENTIALS

Ph.D., Bioinformatics | Leipzig University, Germany | April 2024 | **Magna Cum Laude**
M.Sc., Bioinformatics | Bharathiar University, Tamil Nadu, India | June 2013
B.Sc., Bioinformatics | Kannur University, Kerala, India | June 2011

SELECTED PUBLICATIONS

- **Velluva A**, Radtke M, Horn S, Popp B, *et al.* PTEE resource facilitates tissue selection for RNA-seq-based clinical genetics studies. *BMC Genomics*. 2021.
- **Velluva A**, Le Duc D, Cassatt-Johnstone M, Olsen RA, *et al.* Genomic basis for skin phenotype and cold adaptation in the extinct Steller's sea cow. *Sci Adv*. 2022.
- **Velluva A**, Körner MB, Bundalian L, Radtke M, *et al.* Altered gene expression profiles impair nervous system development in 15q13.3 microdeletion. *Sci Rep.* 2022.
- Barth LAG, Nebe M, Kalwa H, **Velluva A**, *et al.* PLSCR4 regulates adipocyte differentiation via AKT activation. *IJMS.* 2022.
- Le Duc D, Lin CC, Popkova Y, Yang Z, Velluva A, *et al.* Reduced lipolysis in lipoma phenocopies lipid accumulation in obesity. *Int J Obes*. 2021.
- Bundalian L, Su YY, Chen S, **Velluva A**, *et al.* Epilepsies of presumed genetic etiology show enrichment of rare variants occurring in the general population. *Am J Hum Genet.* 2023.